

Astronomy 102 - Lab 07

Light and Spectroscopy

Please have cameras on, microphones muted, ready to begin at 12:05pm

Matthew Prest

The latest in Astronomy

Northern lights may be visible in 18 states tonight!

Alaska

North Dakota

Minnesota

Montana

Washington

Idaho

Wisconsin

South Dakota

Michigan

Maine

Vermont

New Hampshire

Oregon

Wyoming

Iowa

Nebraska

New York (upstate)

Illinois



The latest in Astronomy

What are the Northern (or Southern) Lights? What causes them?



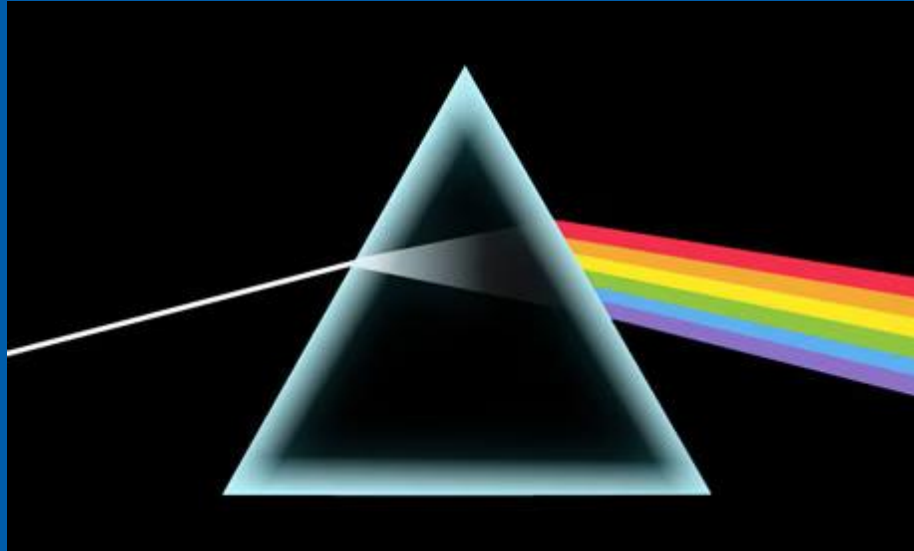
Optics

Optics:

- The study of light and its properties

Spectroscopy:

- The study of how light interacts with matter (stuff)



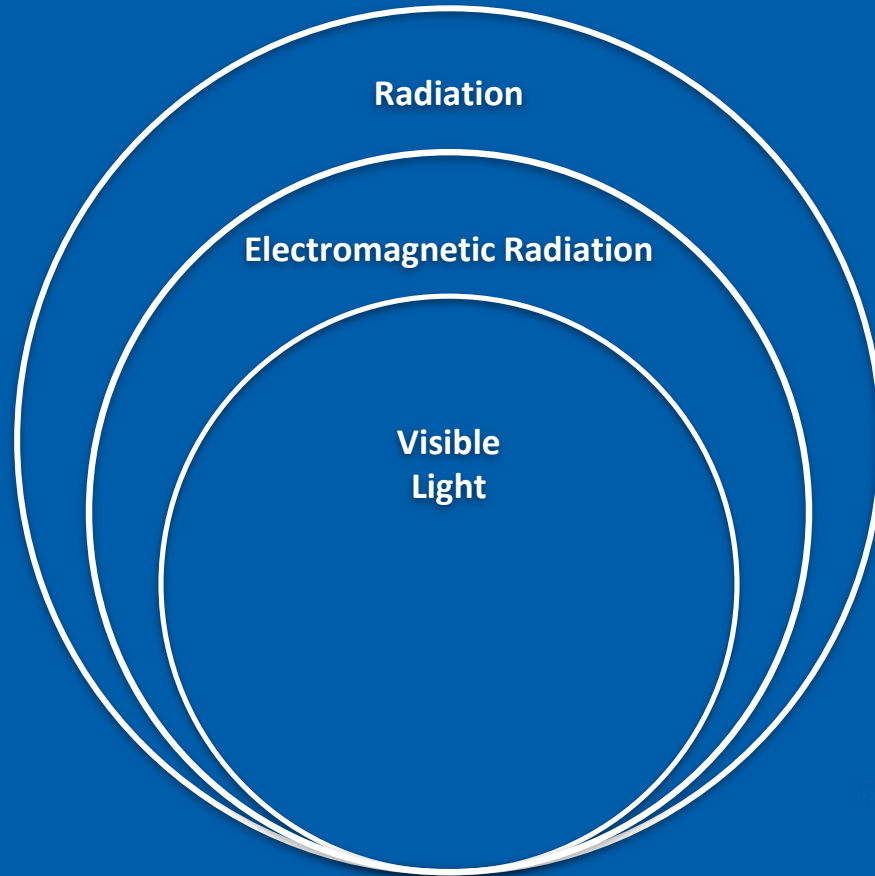
Optics - Background

What is light? What do you think of when you see the word light?

What is radiation? What do you think of when you see the word radiation?

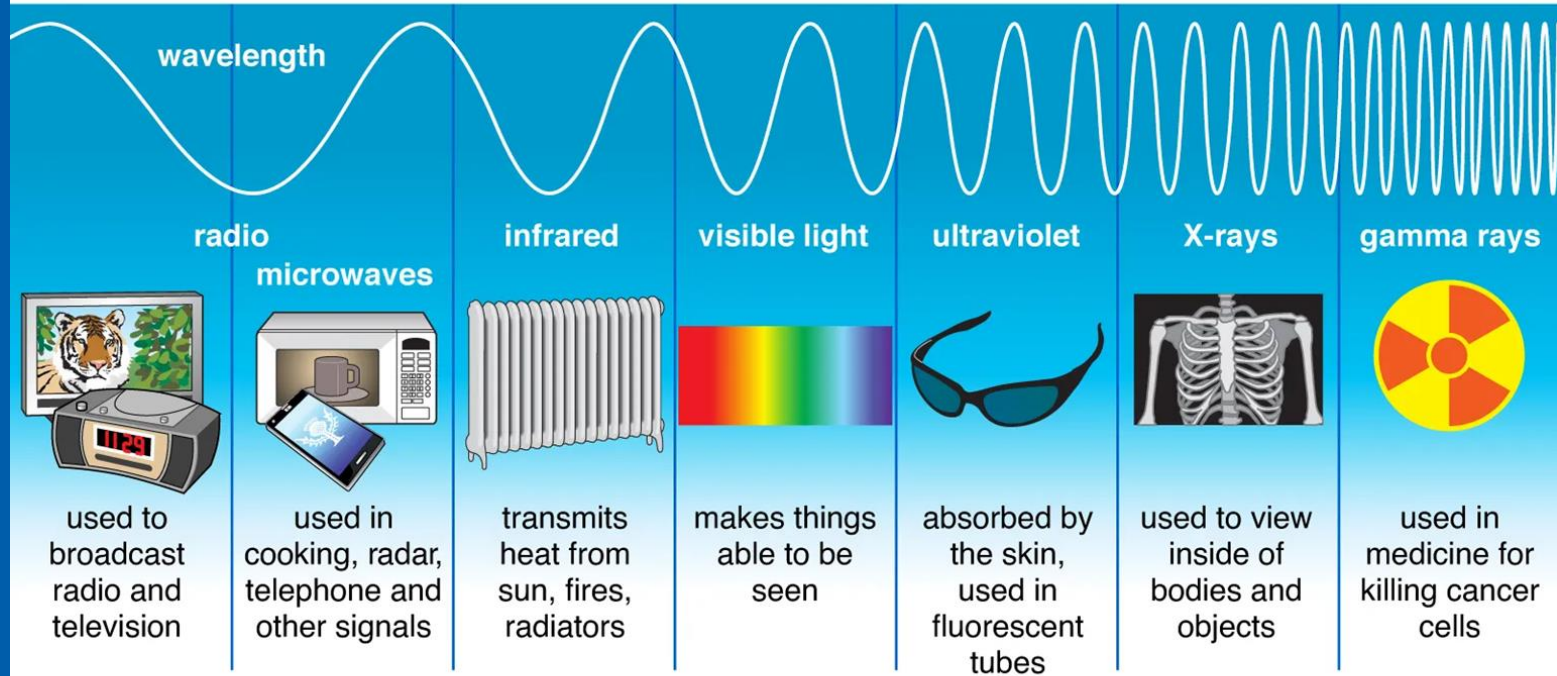


Optics - Background



Optics - Background

Types of Electromagnetic Radiation



Light & Heat

Thermal Radiation

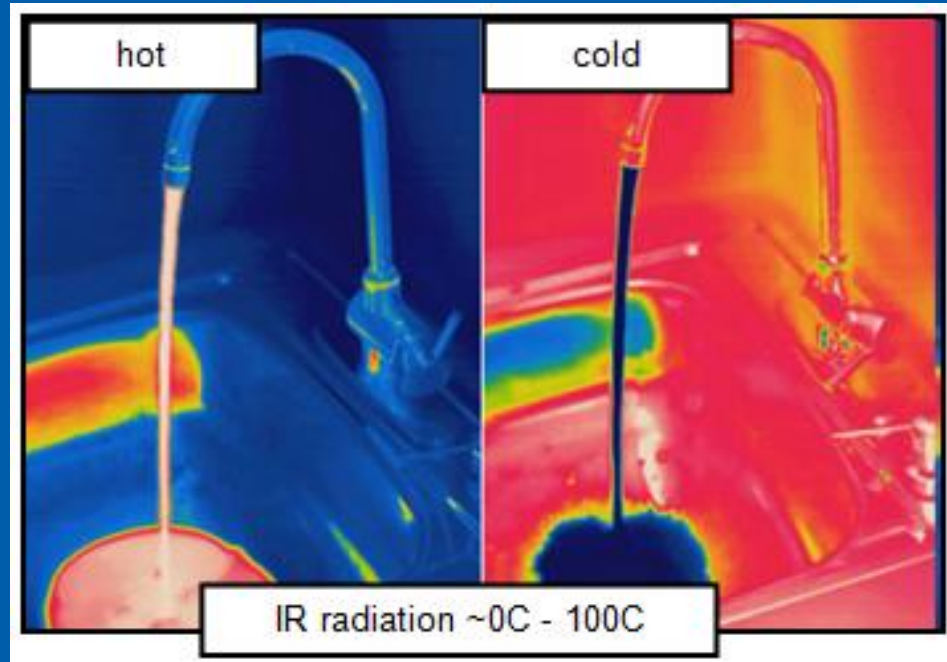
- Why do hot things glow?



Light & Heat

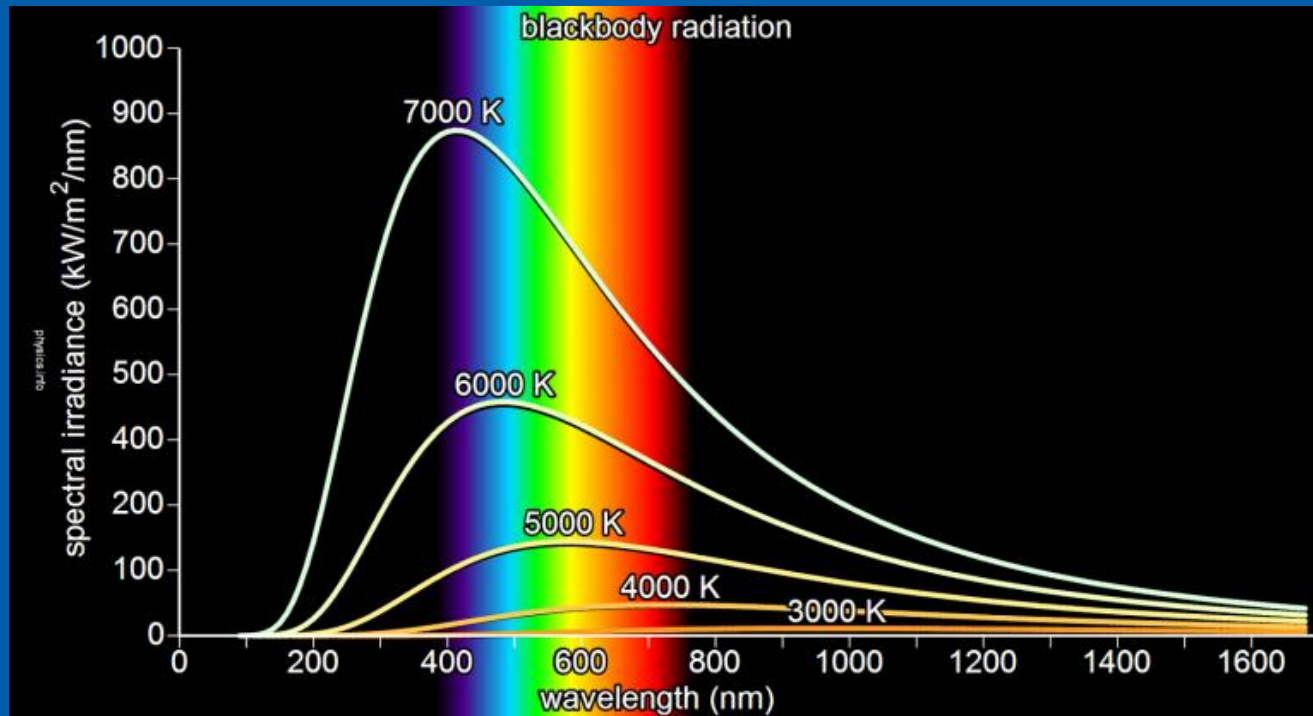
Why do hot things glow?

- Everything is always glowing!
- As we heat the metal up the color of its glow changes to become visible light!



Light & Heat

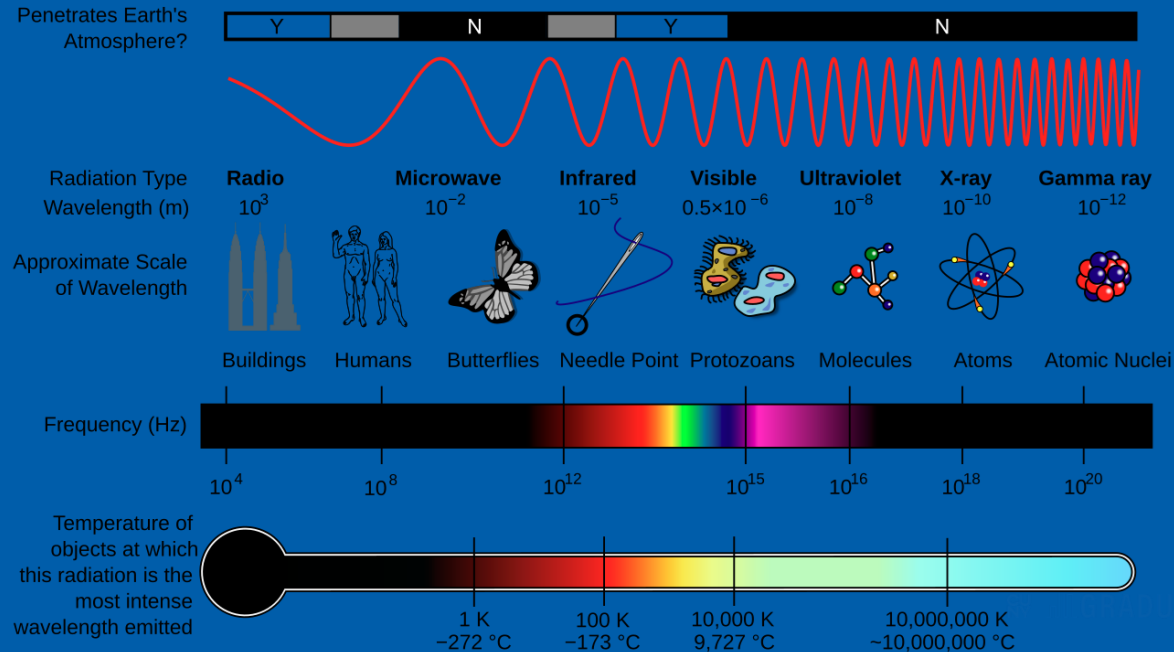
We are in a bath of radiation we just can't always see it



Some Useful Concepts

Wavelength (λ) vs Frequency (f)

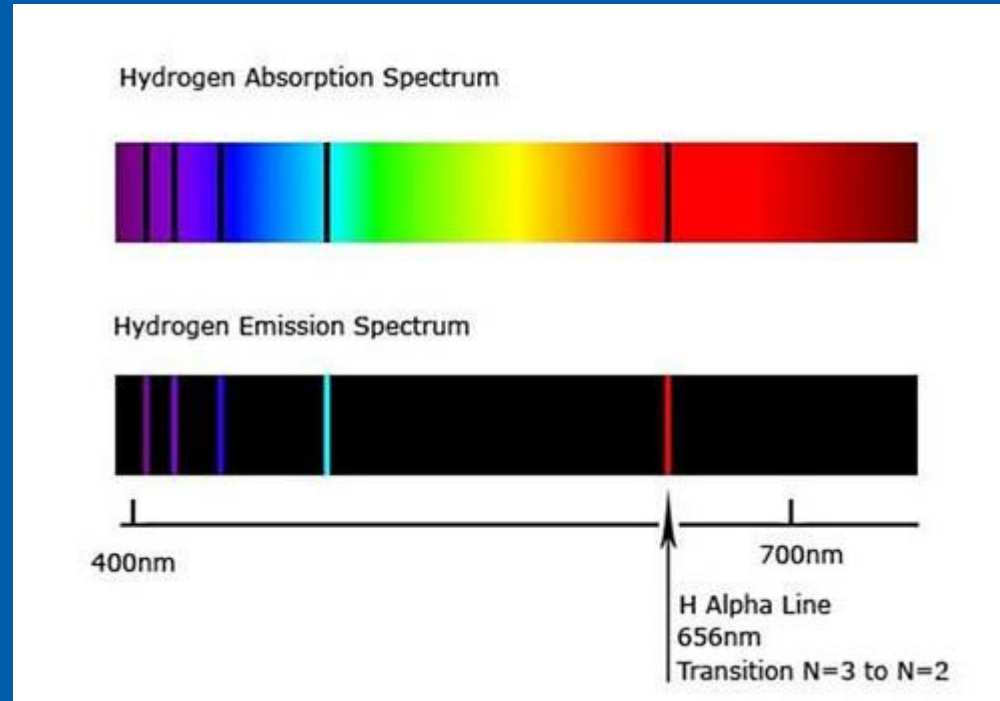
- Like running stride length vs turnover speed
- $c = \lambda f$



Some Useful Concepts

Emission vs Absorption Spectra

- Emission is outgoing radiation
- Absorption is incoming



Reminders

1. Complete the Tutorial first!

- This is best done now in class where I am available to assist

1. Attempt the essay questions last!

- These should be based on what you have covered in the tutorial.

1. Previous week's lab report due at the beginning of the next lab!

- Final grades are based on best 6 online labs plus each of the 4 in-class labs

1. Grades will be posted to Brightspace

- Check now that you have access to our course!
- Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!

1. Office hours

- Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!